DIC26
Cascading Dynamic Circuits

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives and handout requirements

Following handout PDF pages 107–110, deck 01 tests the unsafe same-type cascade in Fig.26.1; deck 02 compares Figs.26.4/26.5 under equal 10 fF coupling and a strength sweep; deck 03 answers Q1 on speed. The three decks contain nine completed analyses without errors or failed measurements. The simulator reports suppressed encrypted-block warning details, which does not establish a warning-free model.

2 Circuits, conditions, and interpretation

The supplied model is evaluated at a declared teaching condition of 1.0 V and 25°C, not a certified voltage rating or reproduction of the original 2.5 V process. Unless stated otherwise, Wn/Wp/L = 0.6/1.8/0.25 µm and output load is 5 fF. The clock period is 10 ns with 20 ps edges. Declared low/high diagnostic thresholds are 0.2/0.8 V; these are not extracted receiver noise margins.

The N-only domino uses two four-transistor dynamic NAND gates with their second inputs tied high, each followed by a static inverter. The N-P chain directly alternates four N/P dynamic stages. Both implement noninverting logic with equal output loads. The N-domino core contains 12 transistors and the N-P core 16. This compares the handout topologies, not equal transistor counts or optimized logical effort.

Every public deck contains only a `MODEL_SETUP_REQUIRED` comment; local model setup is required before execution. Neither model content nor loading paths are distributed. Figures use actual printed data; per-case CSV and measurement JSON files are in `results/`.

3 Step 1: unsafe cascading

Input A rises during precharge. When the clock rises at 2.5 ns, X and Y are initially high, so X initially enables the second stage. Although the first stage later discharges X, lost charge at Y has no active restoration path during evaluation. Y is 0.655339 V at 7 ns, below the 0.8 V high diagnostic threshold. Here the result is substantial high-level degradation rather than a complete discharge to zero; that distinction is retained.

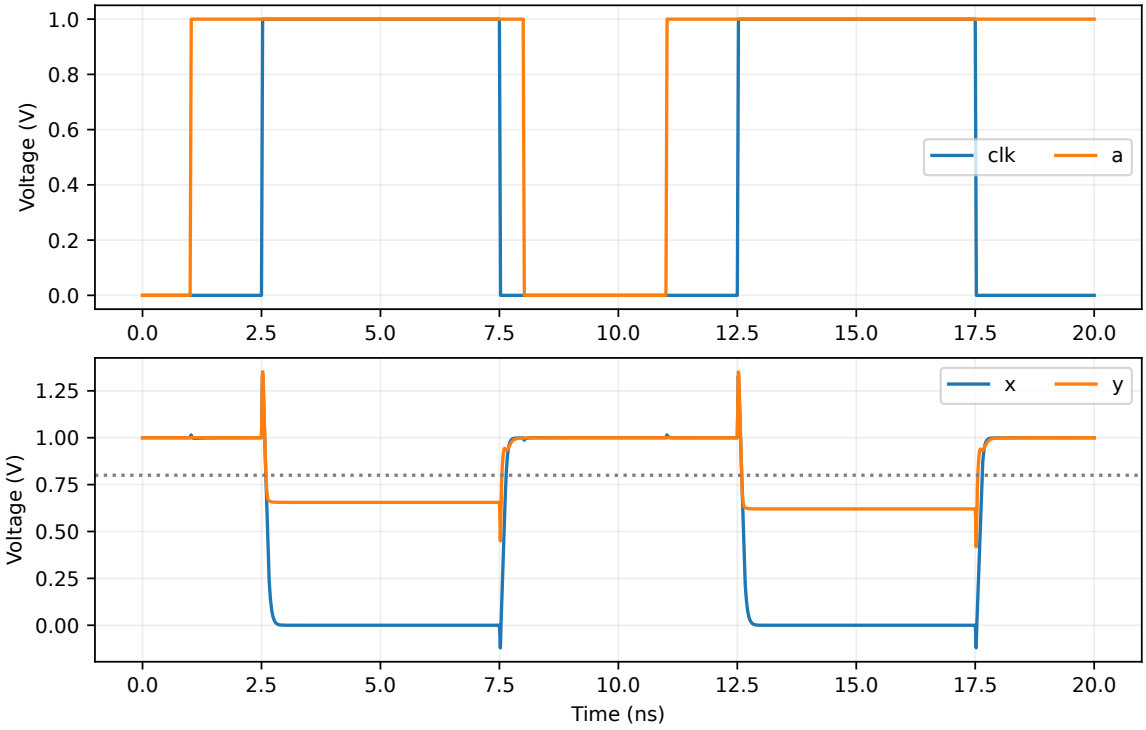


Figure 1: Charge loss in the direct cascade; dotted line is the 0.8 V diagnostic threshold.

4 Step 2: equal-noise comparison

A is fixed at zero, so both outputs should remain low. During evaluation, aggressor Q falls from 1 to 0 V at 4 ns and couples through equal capacitances into each first dynamic node. The handout condition is 10 fF. A static inverter in N-domino rejects small disturbances, whereas N-P stages directly control opposite-type devices and can regenerate smaller floating-node shifts.

C (fF)	N peak (V)	N-P peak (V)
0.001	-0.000011	-0.096783
2	0.000056	-0.096796
5	0.000141	-0.096797
10	0.000249	0.999901
15	0.000443	1.000282
20	1.006744	1.001090
30	1.010412	1.001674

At 10 fF the N-domino peak is only 0.000249 V, while the N-P output reaches approximately 1 V and holds an erroneous high. The distinction persists at 15 fF, whereas both fail at 20/30 fF. The near-zero 0.001 fF control shows N-P clock feedthrough near -0.097 V, which is not clipped to ideal zero. These tests support better N-domino immunity under the stated conditions, not under every sizing, edge, load, or noise waveform.

5 Q1: speed and discussion

A separate input transition at 4 ns during evaluation compares data-path speed without confusing precharge timing with propagation delay. Using 0.5 V input/output crossings, N-domino delay is

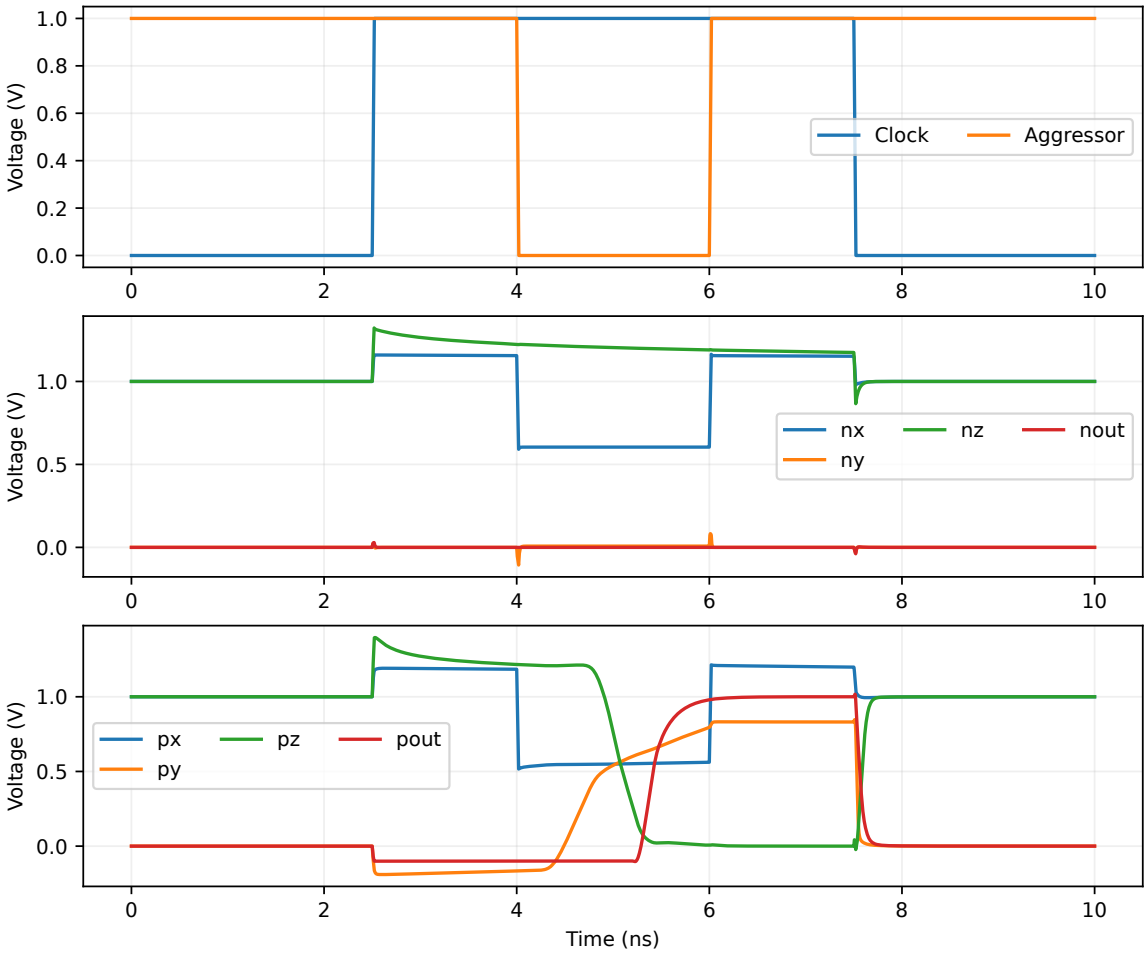


Figure 2: Equal 10 fF disturbances and the internal nodes of each chain.

367.720 ps and N-P delay is 657.328 ps. N-domino is faster in this implementation. Eliminating static inverters does not by itself guarantee a faster N-P chain: PMOS stack drive, node loading, and device sizes also contribute.

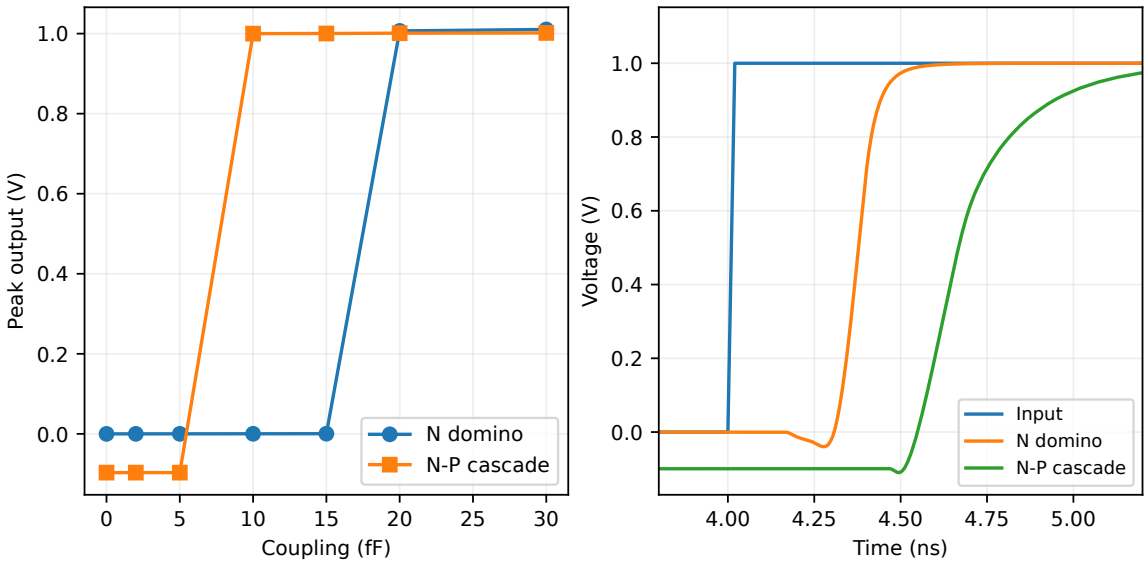


Figure 3: Left: coupling sweep. Right: data-path speed test without coupling capacitors.

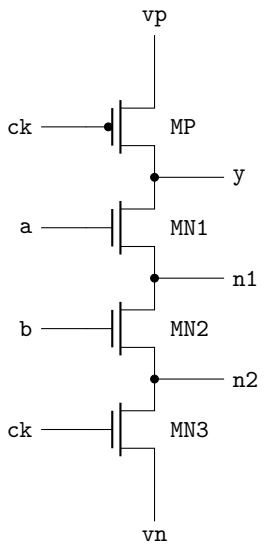
A precharged output must not enable the next evaluation network. Same-type stages therefore need an intervening inverter, while opposite types can connect directly, with analog noise tolerance still requiring evaluation. These are finite circuit simulations, not PVT, post-layout, or complete noise signoff.

6 MOS transistor schematic

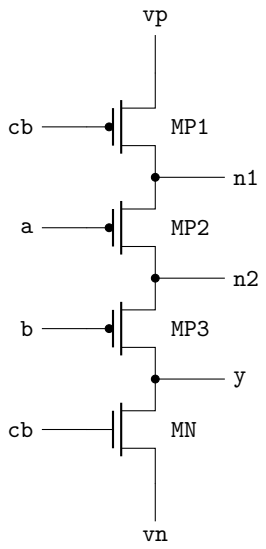
Source netlist: `02_noise_immunity.sp`. N-domino and alternating N/P dynamic chains; the same cells also implement the speed and unsafe-cascade tests.

Identical node labels denote electrical connections. The drawings are editable CircuitikZ source; hierarchical boxes below refer to the MOS cells shown here.

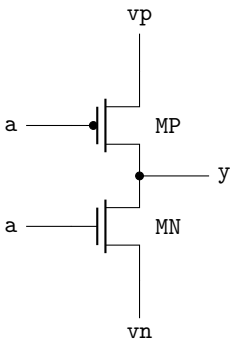
Unless explicitly labeled otherwise, every NMOS body connects to `vn`=0 and every PMOS body to `vp`=V_{DD}; body wires are omitted for readability. Ordinary inverter and dynamic/clocked cells use $W_n/L_n = 0.6/0.25\,\mu$m, $W_p/L_p = 1.8/0.25\,\mu$m. Both TG devices use $W/L = 0.6/0.25\,\mu$m.



`dyn_n(a,b,y,ck,vp,vn)`: 4T



`dyn_p(a,b,y,cb,vp,vn)`: 4T

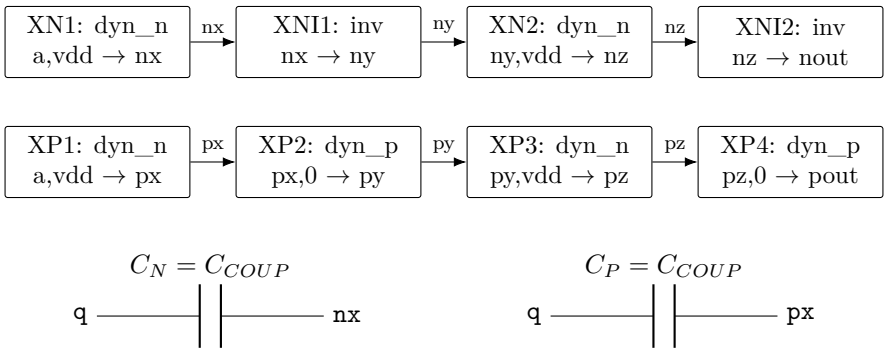


`inv(a,y,vp,vn)`: 2T

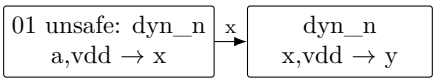
For N cells, the second data input `b` is tied to `vdd`; for P cells it is tied to 0. The PMOS precharge device in the N cell and the NMOS predischarge device in the P cell use the same dimensions stated above.

$V_{DD} = 1$ V; $cb = V_{DD} - V(clk)$.
Clock and data ties.

6.1 Cascades and capacitive disturbance



All N cells use `clk`; all P cells use `cb`. Each output `nout`/`pout` has 5 fF to 0. The two coupling capacitors connect the same aggressor `q` to the first dynamic node of each chain. The base value is 10 fF and the sweeps are 0.001, 2, 5, 15, 20, and 30 fF. For this noise test, `a`=0; `q` falls from 1 V to 0 V at 4 ns.



In `01_unsafe_cascade.sp`, both stages share `clk`; $C_x = C_y = 5$ fF to 0. There is no intervening inverter; a falling first-stage dynamic output directly drives the next evaluation network, whose input should rise during evaluation. The connection shown here is intentionally the unsafe test topology.